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Foundations of Risk Management (APM_5FI11_AE) - Introduction to Banking Risks & Regulation

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Overview of the Commercial / Retail banking business

Retail / commercial banking business – sketch of a balance sheet

Off Balance Sheet

Liquidity Committments (e.g. “autorisation découvert / “overdrafts”)

Guarantees

Assets	Liabilities
Loans (Retail, SME-Small and Medium Enterprises, Corporates, Institutional, Interbank)	**Deposits** **Term Deposits** Retail (natural person) clients, Small and Medium Enterprises, Large Corporate, Financial Institutions
	Regulated Savings Livret A, PEL/CEL, other
Securities Portfolio (Government Bonds, Corporate,...)	**Bond Issuances** EMTN, Convertibles,...
Cash*	**Equity (Capital)**

Credit Exposures

Refinancing scheme (“Liquidity / Funding”)

From a regulatory / legal perspective, we do not speak of a bank, but of a **“credit institution”** whose role is to **take on deposits and grant loans.**

*Cash “at-hand” (except banknotes & coins) usually re-deposed in the Central Bank or in other credit institutions.

Overview of the Corporate and Investment banking (“CIB”) business

Corporate and Investment Banking business (also called wholesale business) – sketch of a balance sheet

Asset	Liabilities
A: Positive value of the hedging portfolios	**B: Negative Market Value (« Fair Value ») of derivative portfolios (C_T)** Swaps, options … Notional values of the derivatives are accounted off-balance sheet
C: Positive Market Value (« Fair Value ») of derivative portfolios Swaps, options … Notional values of the derivatives are accounted off-balance sheet	**D: Negative market value of the derivative hedging portfolios**
Securities trading portfolios	**Bond Issuances** EMTN, Convertibles,…
Cash	**Equity** (shareholders)

Market / Model Values (assets A, C and securities trading portfolios)

Market / Model Values (liabilities B and D)

Refinancing scheme (bond issuances)

From a regulatory / legal perspective, a “credit institution” may run trading activities (under certain constraints).

It is also possible to solely run trading activities under the “Investment Firm” license, which however prevents from taking on deposits and granting loans.

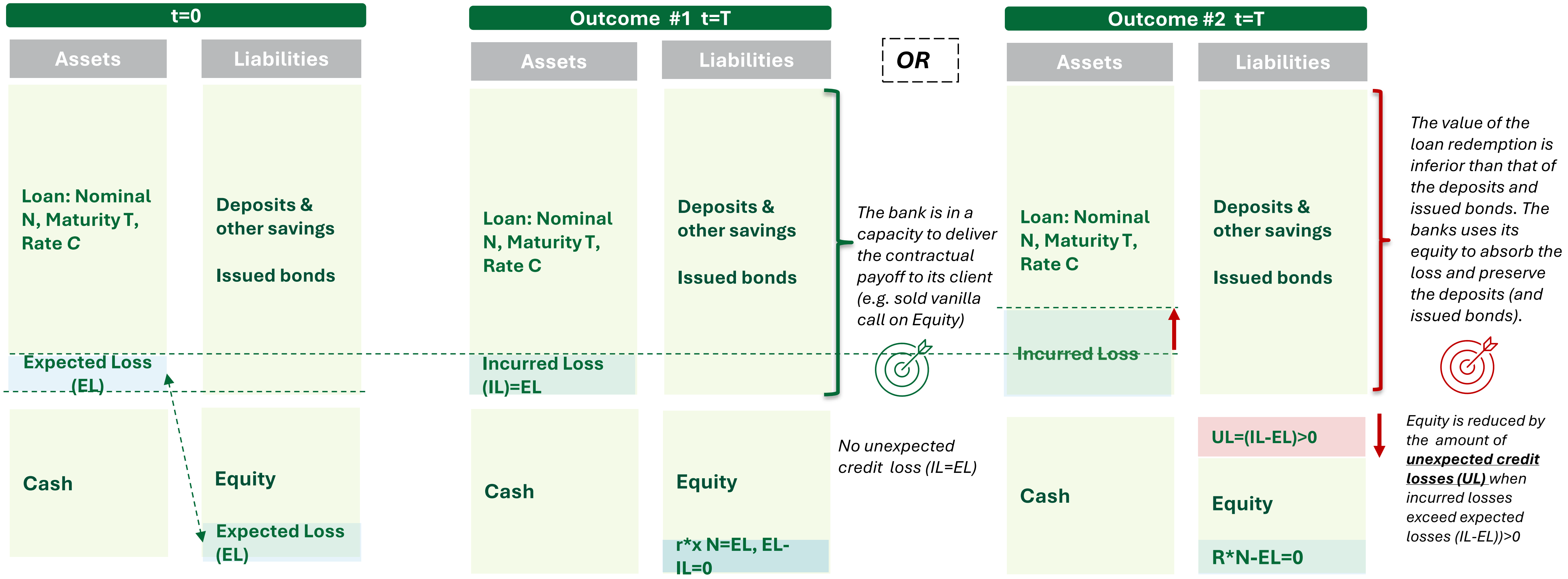
Overview of the Profit & Loss

Illustration of the P&L structure applicable to any banking business activity

Source / Destination	Flow	P&L line	Applicable to ? Retail Bank	Applicable to ? Wholesale Bank (CIB)
From: Loans & Securities	Interest rate income (+)	(1) Net Interest Margin (NII)	Yes	No (or very low)
To: Loans & Securities	Interest rate expense (-)			
From: Client services	Commissions income	(2) Net Commission	Yes	Yes
To: “Market” costs	Commission expense			
From trading business only (CIB)		(3) Net Trading activities gain/loss	No	Yes
		(4)=Net Banking Income (NBI) (1)+(2)+(3)		
		(5) Salaries, Operating Expenses		
Expected Loss & Incurred Loss on Loans & Securities		(6) Cost of Risk	Yes	No
		(7) Operating Income = (4)-(5)-(6)		
His Majesty Treasury (HMT)		Tax		
Bank’s Equity (Shareholders)		Profit & Loss (Net Income)		

Credit Risk - Expected Loss & Unexpected Loss

The **credit risk** is the risk of loss arising from the inability of the borrower to repay part / the totality of the loan nominal and / or interests.



- $\widetilde{V_T}$ the random variable denoting the value of the loan at maturity
- V_T the value of the portfolio at time T in **absence of default** ($N + \sum_{i=1}^{T} N * C$)
- $\tilde{L} = \widetilde{V_T} - V_T$: the loss and $\mathbf{EL} = \mathbb{E}(\tilde{L})$ the expected loss (computed at inception) – **necessitates a model** (second part of the lecture).

- $L^* = N(1-R)$: **incurred loss** (over the life of the loan), R actual recovery.
- $C = r^0 + r^* + r^c$, where $r^* = \frac{EL}{N}$ remunerates **the credit risk**, r^0 remunerates the depositors & bond holders, r^c is the commercial margin.

Market Risk - Expected Loss & Unexpected Loss

The **market risk** is the risk of loss arising from adverse movements in market price affecting the trading business. We present a commonplace illustration related to derivative (option) trading where market price the derivative underlying (e.g. vanilla equity call).

t=0

Assets	Liabilities
Positive Model Value (« Fair Value ») of the Hedging Portfolio: H_0	Negative market value of the (sold) option: C_0
Expected Losses("reserves"): U_0	Expected Losses("reserves): U_0
Cash	Equity
	+ commercial margin M_0

Outcome #1 t=T

Assets	Liabilities
Positive Model /Market Value (« Fair Value ») of the Hedging Portfolio: H_T	Contractual Pay-off of the (sold) option due to the client: C_T=H_T+U_0
Expected Losses ("reserves): U_0	
Cash	Equity
	+ commercial margin M_0

The bank is in a capacity to deliver the contractual payoff to its client (ex: sold vanilla call on Equity)

No unexpected trading loss

OR

Outcome #2 t=T

Assets	Liabilities
Positive Model/Market Value (« Fair Value ») of the Hedging Portfolio: H_T	Contractual Pay-off of the (sold) option due to the client : C_T > H_T+U_0
Expected Losses ("reserves): U_0	
Cash	Equity
	+ commercial margin M_0
- Cash	- Unexpected Trading loss

*The value of the hedging portfolio is lower than the contractual payoff due to its client (ex: sold vanilla call on Equity). The bank must then use its **equity** to deliver the pay off.*

Equity is reduced by the amount of unexpected losses (UL). Equity > UL

How to adequately measure the UL ?

The **reserves** (e.g. model, calibration reserves) **U_0 are paid by the client** to cover all expected model / market calibration uncertainties.

On example #1, C_T=H_T+U_0, namely the hedging strategy and the reserves equal the payoff due to the client.

When C_T > H_T+U_0, the bank delivers the contractual payoff. However it entails an unexpected loss (i.e. a severe loss which rarely occurs) covered by the **shareholders**, hence reducing the bank's Equity accordingly. The available cash enables the bank to pay the amount due to its client.

Structural interest rate risk

- Illustration of the main risks faced by a commercial credit institution (bank) – simple balance sheet made of 1 Loan (fixed rate) refinanced by 1 Bond (variable rate – Euribor). We assume that Euribor: i) suddenly raises by 200 bps, ii) Euribor falls by 200 bps and remain fixed in both cases. Whenever a fixed rate faces a variable rate (customary in any bank business) it entails an **interest rate gap** or structural interest rate risk.
- We analyse, all other things remaining equal, whether the Equity reduces or rises (in accordance with the Net Interest Income –"NII").

t=0

Assets	Liabilities
Loan: Nominal N, bullet amortization Maturity T, Fixed Rate,	**Bond, Nominal N, bullet amortization Maturity T, Variable Rate (BOR)**
Cash	**Equity (t=0)**

Years	1	2	3	4	5
Interest Rate Received (FR) from the Loan	FR	FR	FR	FR	FR
Variable Rate (BOR) paid on the bond	BOR	BOR	BOR	BOR	BOR
Net Interest Income (NII)	(FR-BOR)	(FR-BOR)	(FR-BOR)	(FR-BOR)	(FR-BOR)
Assumption : Euribor + 200 bps = > NII :	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)
Assumption : Euribor - 200 bps = > NII :	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)

Equity (t=0)

Equity t=1 | **Equity t=2** | **Equity t=3** | **Equity t=4** | **Equity t=5**

OR

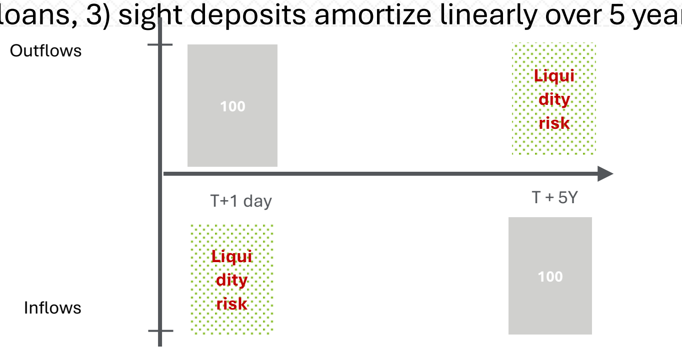
Equity t=1 | **Equity t=2** | **Equity t=3** | **Equity t=4** | **Equity t=5**

- **How is it possible to hedge a fixed rate position refinanced by a variable rate position ?** By using an instruments which transforms a fixed rate into variable rate !
- Regulators impose limits on the growth / reduction of Equity following a stress on the interest rate.

Liquidity Risk – introduction and illustration

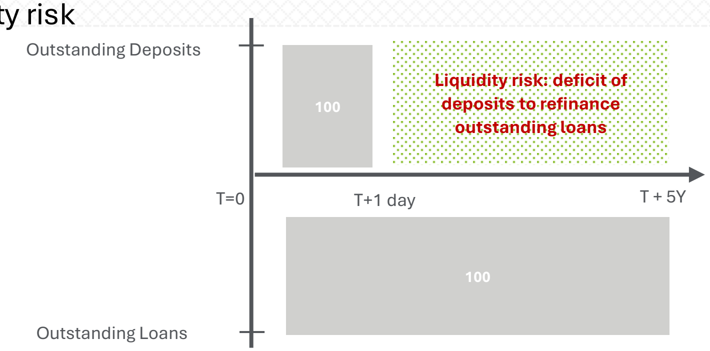
Liquidity risk occurs whenever cash outflows do not match cash inflows. In case a banks fails to repay its debt (i.e. cash outflow) it triggers an immediate bankruptcy. Liquidity risk is the **most critical risk borne by a bank**. Illustration through 3 schemes : 1) sight deposits amortize immediately: immediate liquidity risk , 2) bonds amortize perfectly match the loan amortization of the loans, 3) sight deposits amortize linearly over 5 years: mitigated liquidity risk

Asset	Liabilities
Loans : 100, bullet profile, 5 years	Sight deposits: 100
Cash	Equity

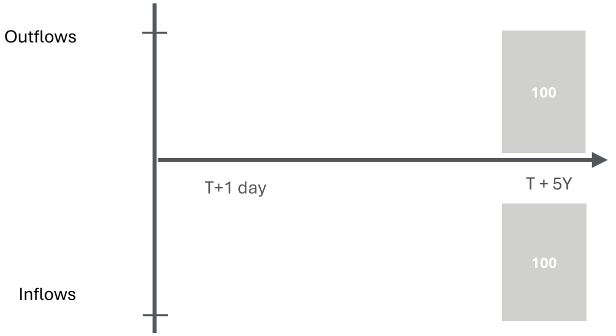


Left: view on the outflows / inflows

Right: equivalent view expressed in outstanding amounts

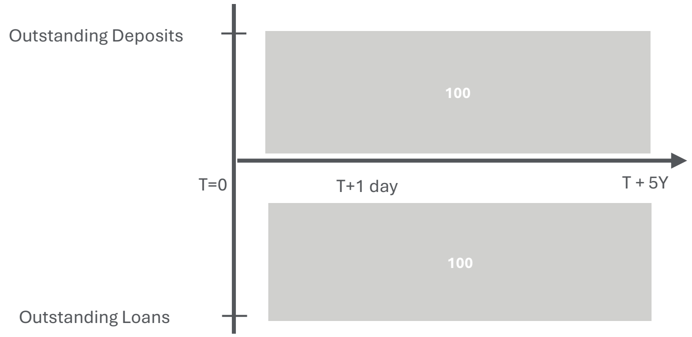


Asset	Liabilities
Loans : 100, bullet profile, 5 years	Bonds: 100, bullet profile, 5 years
Cash	Equity

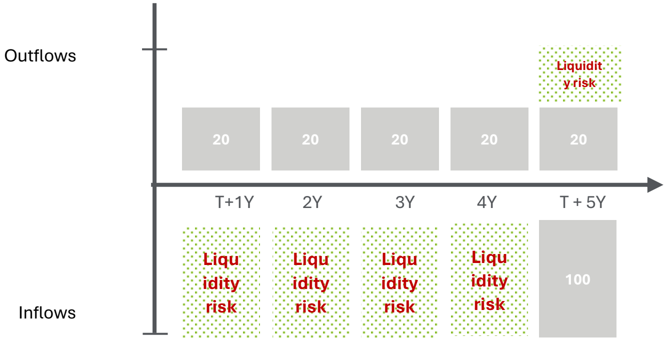


Left: view on the outflows / inflows

Right: equivalent view expressed in outstanding amounts

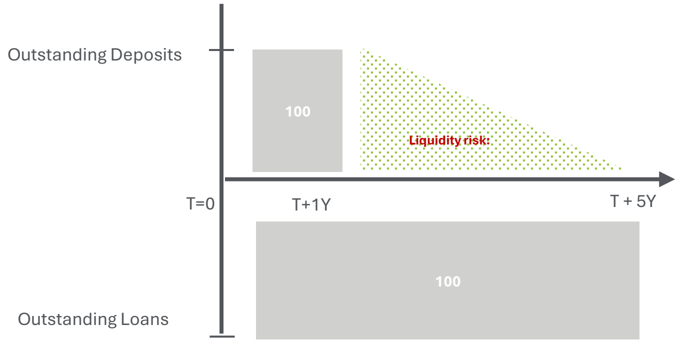


Asset	Liabilities
Loans : 100, bullet profile, 5 years	Sight deposits: 100
Cash	Equity



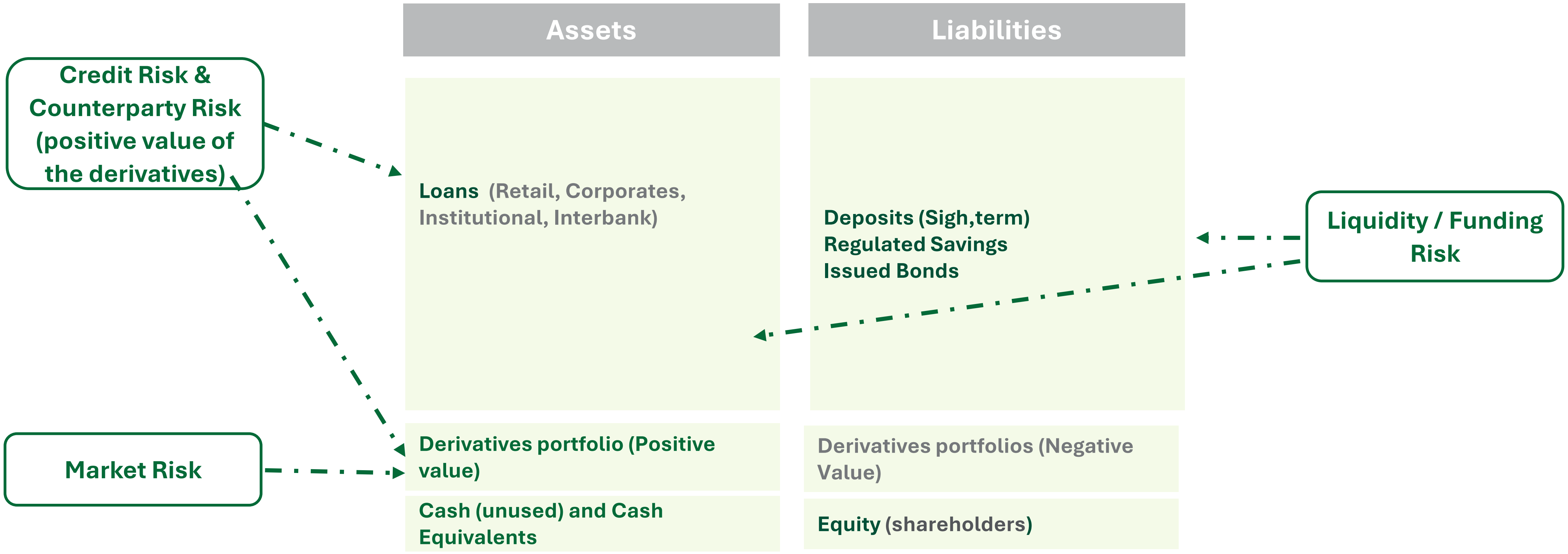
Left: view on the outflows / inflows

Right: equivalent view expressed in outstanding amounts



Summary of the main banking Risks

Illustration of the main risks faced by a commercial credit institution (bank) – sketch of a balance sheet



Summary of the main banking risks

Credit & Market Risk: bank's clients do not repay part / totality of their loans / trading business bears losses. Positive value of derivatives also constitute a credit exposure denoted **"Counterparty Credit Risk"**

Structural interest rate risk: bank's Net Interest Income is subject to variations of the variable interest rates

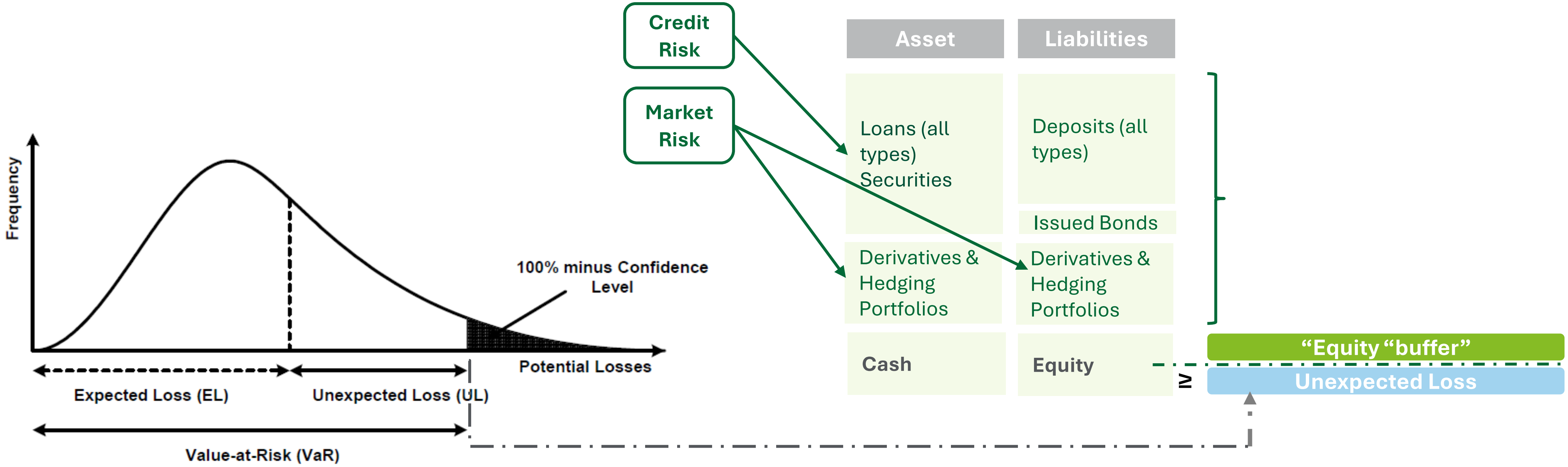
Liquidity / Funding Risks : bank's cash outflows & inflows are not perfectly matched

How to cover the credit and / or market risk(s)?

Illustration of the role of the equity to cover the risks (Commercial, CIB or universal bank and)– sketch of a Balance sheet

Principle: the (regulatory) equity capital must allow the institution to have its shareholders absorb the Unexpected Losses (UL). How to measure the UL and at which time horizon ? **You will see that different risk measures exist (second part of the course).**

So far regulators consider UL occurring in **0.01% (99,9% percentile) of the time over 1-year** (annually equivalent to the failure of 1 institution out of 1000).

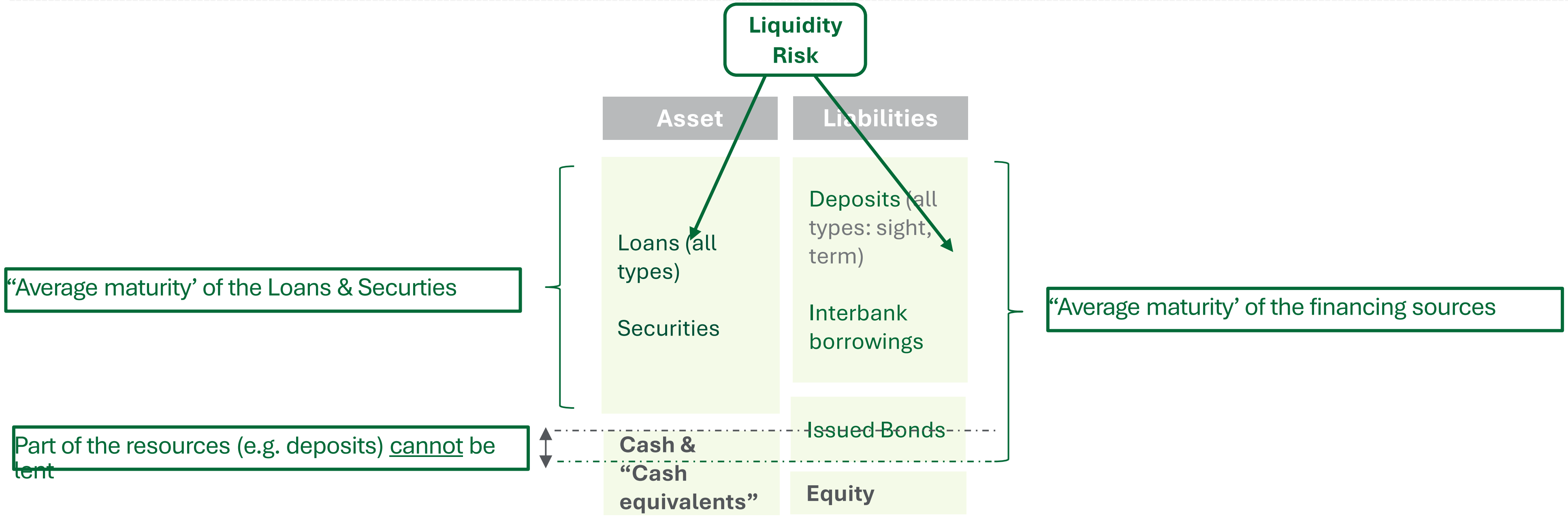


Regulators have streamlined the measurement and requirements for the minimum required amount of Equity (*Pillar 1*) to operate so that all institutions (banks) can comply with these measures and for the initial purpose of comparability (« one size fits all »). **Pillar 2** tailors the minimum required amount of capital through a series of eventually binding adjustments over the Pillar 1 (« bespoke approach »).

How to cover the liquidity risk?

Illustration on the constraint impacting the funding strategy to reduce liquidity risk.

Liquidity risk may be covered by setting additional constraints to ensure that: i) the maturity of the assets do not exceed "on average" those of the liabilities, ii) the bank is in a capacity to withstand a bank run over a short timeframe.



Regulators have streamlined the liquidity & funding constraints under a sheer "accounting based approach" applicable to all banks (Pillar 1 – "one size fits all").

Banking regulation: the 3 Pillars

Based on 3 pillars, the Basel Committee (and thereby any country adopting the Basel Standards) has streamlined the stability of the banking system

Pillar 1: Minimum Requirements (“one-size fits all”) to operate

- Solvency ratio
- Short-term liquidity ratio (LCR)
- LT Liquidity Ratio (NSFR)
- Leverage ratio

Pillar 2: Supervisory Review Process (introduced on the surface in this lecture)

- “Bespoke” Capital measurement above minimum requirements (Pillar 1)
- Internal Capital / Liquidity Adequacy Assessment Process (“ICAAP” and “ILAAP”)

Pillar 3: Market discipline (not addressed in this lecture)

- Banks are required to disclose numerous and granular information related to Pillar 1, and Pillar 2 (not always) and more generally to their risk management practices and detail on regulatory capital.

Legal regulatory framework prevailing in the EU and streamlining banking risks

Overview of the main regulation and directives for regulatory reporting

Legal Rules in force in the EU

Generic

Basel I,II,III, IV (upcoming) agreements
Who? Basel Committee

→ NOT LEGALLY BINDING UNTIL TRANSLATED INTO THE LEGAL APPLICABLE FRAMEWORK PREVAILING FOR EACH JURISDICTION

CRR I, CRR II, CRR III (2025*), CRD IV & CRD V
Who ? (European Commission, Parliament and Counsel)

Specific

Commission Implementing Regulation (CIR)
Who ? European Commission

Regulatory Technical Standards (RTS)
Who ? European Commission or European Banking Authority (EBA)

Reporting standards

Implementing Technical Standards (ITS)

** EXCEPT FOR THE FUNDAMENTAL REFORM OF THE TRADING BOOK ENTERING IN FOR IN 2026*

Capital (Regulation and Directive) Requirements

CRR I: EU 2013/575

CRR II: EU 2019/876

CRR III: EU 2024/1623

CRD IV: 2013/36 EU

CRD V[1] : EU 2019/878

CRD VI: EU 2024/1619

Be mindful that the CRR II amends /completes (**but not supersedes**) the CRR I and hence the two documents **should be read in parallel (the CRR II does not recite the CRR I)**

Links to consolidated versions

All consolidated Regulation versions: here

All consolidated Directive versions: here

1: TRANSLATED INTO FRENCH LAW BY THE « ARRÊTÉ DU 3 NOVEMBRE 2014 » (CONTRÔLE INTERNE, SURVEILLANCE PRUDENTIELLE (PILIER2) AND COMPLETED BY THE « ARRÊTÉ DU 25 FÉVRIER 2021 » (CONTRÔLE INTERNE)

Pillar 1: Minimum required capital (equity) – introduction to the Solvency Ratio

TOTAL CAPITAL (EQUITY + « EQUITY LIKE »)**

$\geq 8\%$

Risk Weighted Assets (RWA*) = (A+B+C+D+E) – EL (Credit & Market Risk – as already reducing the Equity – no double penalty)

	A Credit risk	B Operational risk	C Market risk	D Counterparty risk – Default risk counterparty	D Counterparty risk – CVA risk*	E Other complements*
Non complex banks	Standard approach	Business Indicator ("BI") approach	Standard approach(es) (and FRTB)	SA-CCR method Simplified SA-CCR method	Basic approach (BA-CVA)	Settlement and delivery risk Securitization
Complex banks	Internal rating model (IRB) & output floor		Internal model approaches FRTB* (1st Jan 2027)	OEM method Internal model (IMM)	Standard approach (SA-CVA) – grounded on IMM	

We will see that this amount of 8% constitutes a bare minimum to which additional binding & less binding constraints are added.

As the measurement of the Unexpected Loss (UL) cannot be left at the hand of each bank as it would be difficult for a supervisory body to ensure that the computation is correct and would also impede comparison between banks solvency. Therefore, regulators impose to all banks (institutions) a "ready-to-wear / one size fits all" minimum amount of regulatory capital (Pillar 1) under the form of a Solvency Ratio. This minimum amount of capital should be at least equal to 8% of the Risk Weighted assets for Credit, Op Risk, Market, Counterparty Risk, Other. 2 broad sets of methodologies are available to compute the RWA (or sometimes directly the Minimum Capital Amount): i) "arbitrary method" dedicated to all "non-complex" institutions, mostly based on accounting figures and easy to implement, but requiring a high level of capital ii) internal model method(s), based on statistical methods, complex and costly to implement & maintain but requiring a lesser level of capital.

The spirit of these 2 methodologies aim at setting the minimum level of capital to withstand an (unexpected) loss at 1 year time horizon with a quantile of 99,9% (i.e. occurring 0.1% of the time).

*not addressed in this course

** we will address at the end of the document some specificities related to Equity and how regulatory capital encapsulates the notion of Shareholder's Equity we addressed previously.

Pillar 1: A - Credit Risk Measurement (overview)

Standard approach

Description *(CRR. Article 107)*

For the purpose of calculating risk-weighted exposure amounts, (arbitrary) risk weights (RW) provided by the are applied to all exposures unless they are deducted from capital.

The determination of the category, the credit quality and the weighting applied are standard and predetermined by the regulatory requirements.

- ***RWA = EAD * RW$_{(SA)}$***

The EADs **(exposures at default)**, excluding derivatives, are determined from the **accounting values** which are :

i. Either substituted by the value of the mortgaged property (residential or commercial) within the framework of mortgage loans and by applying a specific weighting (RW) on the mortgaged property.
   - **Ex:** a mortgage loan of 100 on a retail client with a mortgage valued at 60 gives rise to an EAD of 40 (100-60) on the retail client (RW=75%) and an EAD of 60 on the mortgaged property (RW = 35%), i.e. a total RWA = 40*75%+60*35%=51 and a minimum of amount of equity (capital) = 8%*51=4.08

ii. Or (if applicable) reduced by the value of collateral (whose value is itself mitigated according to criteria specified in the regulation) according to the methodologies set out in the so-called Credit Risk Mitigation (“CRM”) techniques

EADs (exposures) **for derivatives** are determined using the SA-CCR, OEM or IMM methodologies (discussed below).

The regulatory weights (***RW$_{(SA)}$***) depends on the category of the client and are (arbitrarily & conservatively) set by the regulatory rules (Retail, Corporate, Govies, etc.) and, if applicable, its rating. For example, the weightings (RW) applicable to exposures to EU Govies are zero, those for retail clients 75%.c

Internal rating approach (IRB)

Description *(CRR. Article 142 to 191)*

A "rating system" is the set of methods, processes, controls, data collection systems and information systems that have been developed for a given type of exposure and that allow for the assessment of credit risk, the assignment of exposures to a given tier or category and the quantification of default probability and loss estimates.

The use of this approach requires prior validation from the supervisory bodies and prerequisites in terms of rating systems and risk quantification.

$$RW = \left(LGD \cdot N\left(\frac{1}{\sqrt{1-R}} \cdot G(PD) + \sqrt{\frac{R}{1-R}}\, G(0.999) \right) - LGD \cdot PD \right) \cdot 12{,}5 \cdot 1{,}06$$

Weighted exposure amounts will be determined:

- ***RWA = EAD * RW$_{(IRB)}$***
- ***RW$_{(IRB)}$ = f(PD,LGD) - "Credit VaR" formula;***
- ("G" is inverse the distribution function of the Normal distribution, and "R" is the regulatory correlation coefficient). **The Credit VaR formula will be derived in the second part of the lecture.**

Th aches IRB Foundation – IRB Advanced

IRB Foundation	**IRB Advanced**		
Internal model	**Internal model**	PD (Probability of default)	Principle of uniqueness of the PD associated with a counterparty
Regulatory	**Internal model**	LGD (Loss Given Default Rate)	•Economic definition of losses: capital losses + interest, management and carrying costs,... •Whilst the PD is associated with a counterparty, the LGD is associated with a transaction
Regulatory	**Internal model**	EAD (exposure at default)	•Balance sheet: amount drawn (less collateral mitigating the exposure) •Off-balance sheet: estimated exposure at time of default
Regulatory	**Internal model**	M (Maturity)	Residual duration

Pillar 1: B – Operational Risk Measurement (overview) – CRR 3

Not for the final exam – solely for information purpose

CRR 3 abolishes all previous approaches and adopts only a new standard approach

Operational risk - Pillar 1 capital requirements - Business Indicator Component (BIC) - Articles 312-315	**ENTRY IN FORCE: 30 June 2026(*)**

$$\text{Pillar 1 Capital} = BIC = \begin{cases} 0,12 \times BI, & \text{if } BI \leq 1 \\ 0,12 + 0,15 \times (BI - 1), & \text{if } 1 < BI \leq 30 \\ 4,47 + 0,18 \times (BI - 30), & \text{if } BI > 30 \end{cases}$$

The business indicator (BI) is expressed in billions of Euros. It is the sum of 3 components: the interest, leases and dividends component (ILDC), the service component (SC) and the financial component (FC). The EBA must specify a certain number of elements enabling the calculations below to be made.

Business Indicator

The **Interest, Leases and Dividends Component** (**IDLC**) is determined by a formula based on interest income and expenses, assets (outstanding loans, advances and interest-bearing securities) and dividend income. *All components are averaged over 3 years.* ***Interest Component (IC)*** *corresponds largely to NIM from leasing activities (financial and operational).* ***Asset Component (AC)*** *corresponds to the amount of exposures (loans, bonds, etc.) and* ***Dividend Component (DC)*** *from dividends from portfolios, equity funds and non-consolidated holdings.*

$$IDLC = min(IC, 0,0225 \times AC) + DC$$

The **Service Component** (**SC**) is determined on the basis of operating income and expenses from banking operations **(OI, OE)**, commission income and expenses **(FI, FE)** from advisory and service activities (including outsourced services for which the institution is the delegator or delegatee). *All components are averaged over 3 years.*

$$SC = max(OI, OE) + max(FI, FE)$$

The **Financial Component (FC)** is determined from the P&L of the trading book (**TC**) and the MNI of the banking book (**BC**). *All components are averaged over 3 years.*

$$FC = TC + BC$$

Operational risk: article 323 clarifies the requirements for operational risk management systems. Articles 316 to 322 should not apply to institution displaying a BIC < €750m.

(*) Source CRR : EBA roadmap on strengthening the prudential framework (Annex)

Pillar 1: C – Market Risk Measurement (overview) - current framework

Not for the final exam – solely for information purpose

Components of market risk

- Interest rate risk
  - General risk
  - Specific Risk
- Security risk
  - General risk
  - Specific Risk
- Foreign exchange risk
- Commodity risk

Standard approach

Description *(CRR. Title IV Chapter 2 - 4)*

For the purpose of calculating capital requirements for the various components of market risk, the standardized **approach is based upon a multiplication of accounting-based measures (or simple sensitivities – such as duration) times a regulatory coefficient set (arbitrarily) by the regulatory text.**

Interest rate risk	Security risk	Foreign exchange risk	Commodity risk
• General risk ○ Maturity method: all positions are weighted according to their maturity ○ Duration method: weighted position based on duration • Specific risk: requirement level depending on category, credit quality and residual maturity	• General risk The general risk capital requirement is equal to the overall net position of the institution multiplied by 8% • Specific risk To calculate its specific risk capital requirement, the institution multiplies its overall gross position by 8%.	The capital requirement for currency risk is equal to the sum of the institution's overall net foreign currency position and net gold position in the currency of the reports, multiplied by 8%.	○ Simplified method The capital requirement is the sum of the following items: - 15% of the net long or short position in each commodity; - 3% of gross long and short positions in each commodity. ○ Maturity Table Method: The capital requirement for each commodity is equal to the sum of the netted positions per weighted maturity band, the net residual position after netting carried forward to the next weighted maturity band, and the final weighted residual unmatched position

Internal model approach

Description *(CRR. Title IV - Chapter 5)*

Institutions may calculate their capital requirements for one or more of the above-mentioned risk categories by applying their internal models, instead of the standard methods, after verifying that they comply with the regulatory requirements and obtaining the authorization of the supervisor.

The amounts of **capital requirements** are equal to :

- ***Capital requirements = VaR + Stressed VaR***

In addition, specific additional requirements apply: i) on securities portfolios subject to credit rating migration risk (Incremental Risk Charge **- IRC**), ii) on correlation portfolios (CDO Index & Bespoke) - **Comprehensive Risk Measure**. These two last requirements are based on a VaR measure measured on specific parameters (e.g. correlation parameter on CDOs).

Pillar 1: C – Market Risk Measurement (overview) – upcoming framework (FRTB) - 2027

Standard – advanced approach - FRTB

Description (CRR 325 and al)

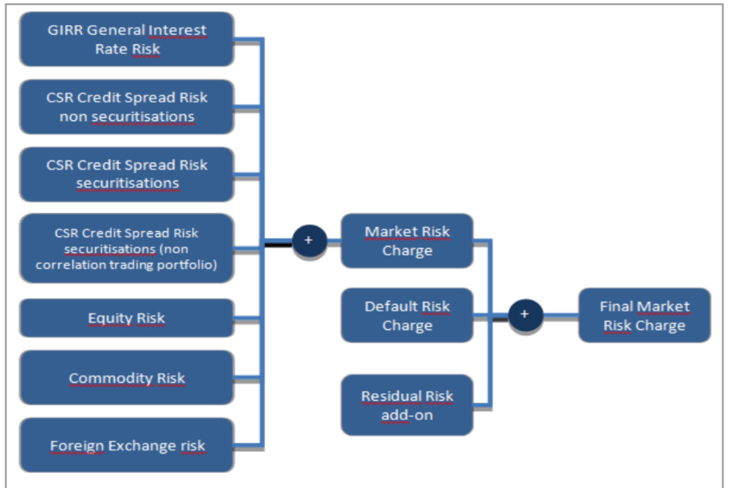
The standard advanced approach will replace the current standard approach (see previous slide) and consists in calculating a sum of capital requirements on 3 dimensions:

- **i)** **Sensitivities** on all asset classes, which are themselves broken down according to :
  - a) Interest rate risk
  - b) Credit spread risk
  - c) Equity, commodity and FX risks.
- ii) Default risk (underlying Equity & Credit)
- iii) The residual risk.

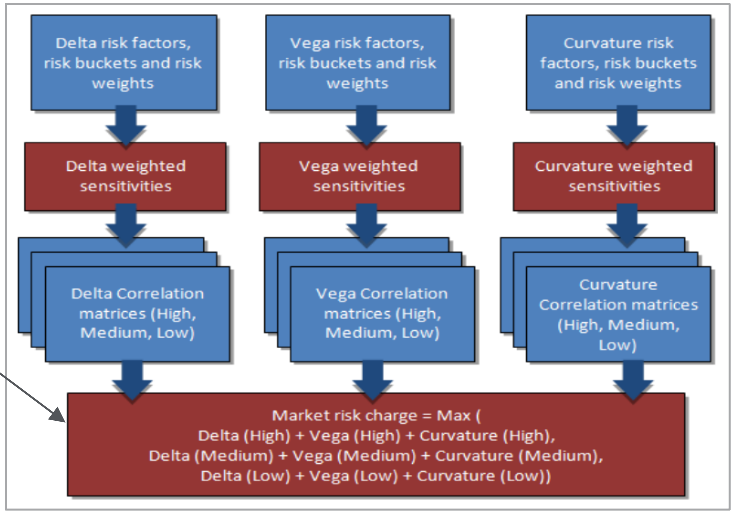
Focus on sensitivities: consists in calculating the capital requirement for market risk by aggregating the three following capital requirements:

- i) Delta risk
- ii) Vega risk (of an option)
- iii) Curvature Risk

These sensitivities are multiplied by (arbitrary) regulatory coefficients and added together (see opposite) to derive the capital requirements.



Calcul du risque final



«Alternative internal model» approach (FRTB)

Description (CRR 325 and al)

The application date for capital requirements is expected in the coming years (2025?).

The weighted requirement amounts will be determined based on **the Expected Shortfall (**which replaces the current VaR) and **a stress scenario**.

=> You will come back to this notion of Expected Shortfall (namely the Expected Loss conditional to **the 97.5%** - quantile) – even more conservative than the VaR.

In addition, an additional requirement for **sudden default risk** will be added on the Equity and Credit portfolios. This calculation will be based on a VaR approach and credit risk hedging instruments may be recognized to mitigate capital requirements.

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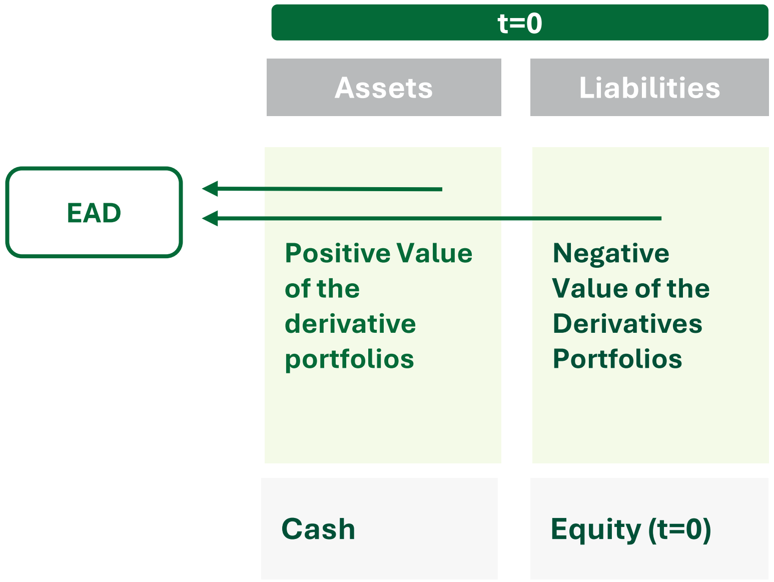
Pillar 1: D – Counterparty Credit Risk

Institutions shall calculate the credit risk stemming from **exposure value** of **derivatives contracts** using one of the methods set out below. As opposed to Loans, derivative value vary overtime in accordance with market conditions (and even pivot from negative to positive and reverse). Henceforth, measuring their exposure EAD is difficult. 4 methods are available : i) two SA-CCR methods and the OEM method aim at calculating the exposure value (EAD) in some specific ways and an Internal Model Method (“IMM”).
The 3 **“non-IMM”** methods are based on the same starting formula: $EAD = \alpha * (Replacement\ cost\ (RC) + Potential\ Future\ Exposure(PFE))$. The IMM computed directly the EAD.
The EAD is multiplied by the corresponding regulatory RW (see A – standardised approach for credit risk) or input into the Credit VaR formula to come up with the corresponding capital requirements.



Original Exposure Method ("OEM")

Conditions of application: the total balance sheet and off-balance sheet of derivative activities must be simultaneously:
(i) ≤ 100 million euros
(ii) ≤ 5% of the institution's total assets

The exposure value (EAD) is calculated separately for each netting set as follows:

Listed and OTC derivatives : $RC = TH + MTA$

Where: i) TH corresponds to the minimum threshold below which the margin call is not authorized, ii) MTA: Minimum Transfer Amount.

Other derivatives : $RC = Max(CMV, 0)$
Where CMV is the current market value of the derivative.

The PFE is equal to the notional amount of the derivative, multiplied by a fixed coefficient that depends on the asset class and the residual maturity. **Ex: for interest rate derivatives:** 0.5%*Residual maturity; 32% for equity derivatives.

2 SA-CCR methods (regular & simplified)

The approach is based on the Market Value of the derivatives, the margin calls, their asset class, notional and does not require performing any simulation.

Conditions for the application of the simplified SA-CCR approach: the total balance sheet and off-balance sheet of derivative activities must be simultaneously.

(i) ≤ 300 million euros
(ii) ≤ 10% of the institution's total assets

Internal model method (IMM)

The approach is based on an **internal model** (**simulation of the of the positive derivatives values** (called « Expected positive exposures ») including margin calls of the bank which must meet quantitative, qualitative and data requirements.

Necessitates the validation from the supervisor (e.g. ACPR in France or ECB in Europe.

Banking regulation: the notion of (regulatory) capital – capital requirements (1/2)

Institutions are required to strengthen their capital in addition to the Minimum Pillar 1. All other things remaining equal, it reduces the profitability (Return on Equity "RoE" diminishes).

	Minimum Pillar 1 Capital requirements			
Common Equity Tier 1 (CET 1) corresponds to the Equity (i.e. Shareholder's (bank's owners) money + accumulated results over the years)	Common Equity Tier 1 (« CET 1 ») (Shareholder's Equty)	***At least**** **4.5% of RWA**	**TIER 1 CAPITAL (*at least** 6% of RWA)**	**TOTAL CAPITAL ("TC")** *At least** 8% of RWA**
Debt instrument (does not entail any ownership) whose features are similar to those of a share (e.g. no redemption date, subordination, discretionary coupon payment)	Additional Tier 1 (« AT1 »)			
Debt instrument (does not entail any ownership) whose features are similar to those of a share (e.g. no redemption date, subordination, discretionary coupon payment)	Tier 2 (« T2 »)		**TIER 2 CAPITAL**	
Additional layer to the CET 1 to ensure that banks do not manage their business too close from the Minimum CET 1 – supervisor have the power to limit dividend distribution to restore CCB.	Capital conservation buffer (CCB)	**2.5% of Common Equity Tier 1**		
Additional layer to the CET1 built during "bullish" economic cycle and which could be used during "bearish" economic cycles (e.g. in the EU, supervisor set the CCyB to 0%.	Counter-cyclical buffer (CCyB)	**Between 0 and 2.5% of Common Equity Tier 1**		
Additional layer to the CET1 for the Globally Systemic Important Institutions (total balance sheet at least above € 30 Bio).	G-SII requirements	**Between 0 and 3.5% of Common Equity Tier 1**		

**See Pillar 2 "add-ons"*
***Also called SREP requirements for ECB directly supervised institutions (banksà*

Banking regulation: the notion of (regulatory) capital – capital requirements (

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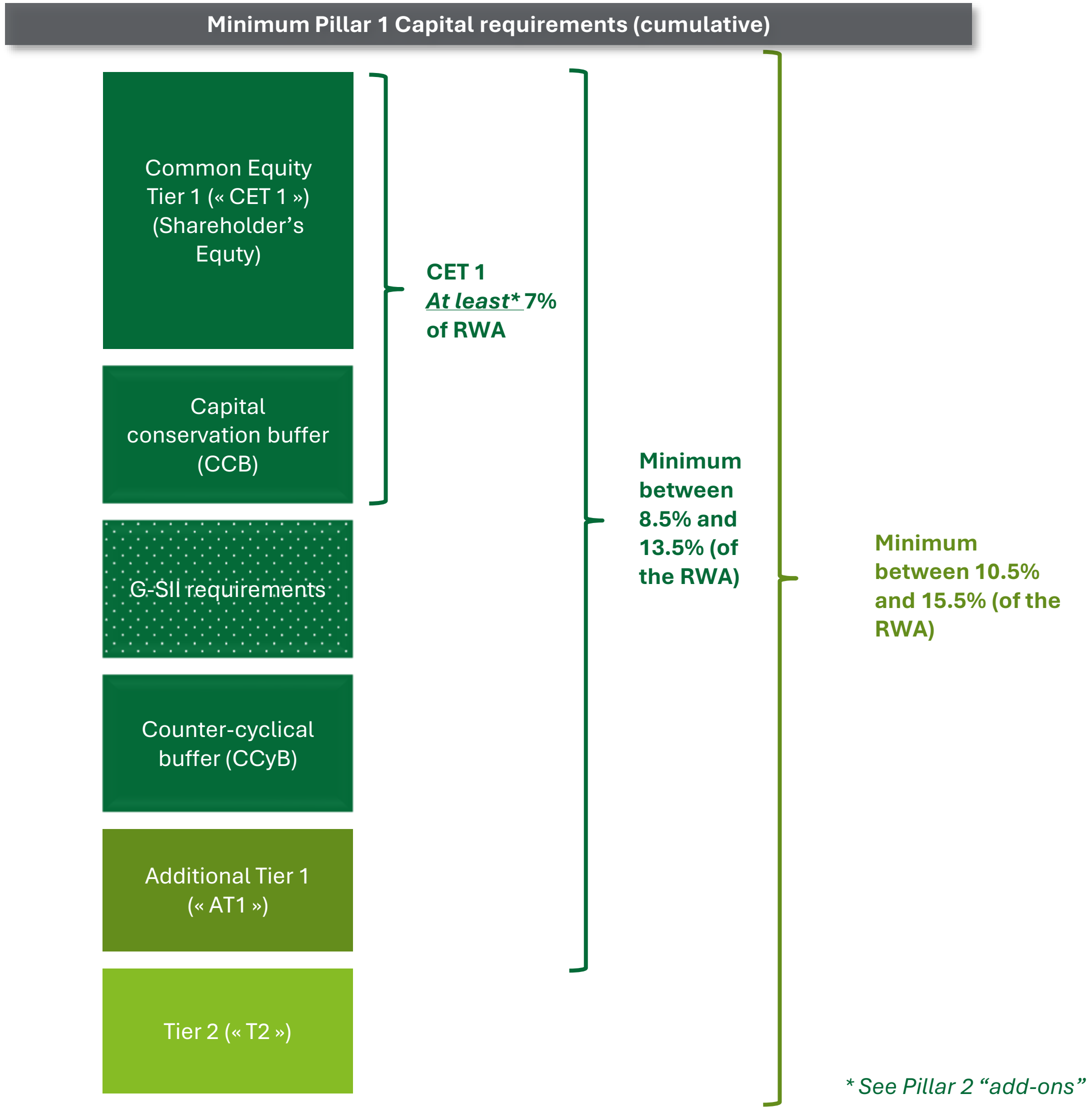
Institutions are required to strengthen their capital in addition to the Minimum Pillar 1. All other things remaining equal, it reduces the profitability (Return on Equity "RoE" diminishes).



Pillar 1: Minimum required capital (equity) – introduction to the Leverage Ratio

The leverage ratio does not embed any risk component (i.e. no statistical measure or unexpected loss concept) and is purely accounting based to facilitate comparison between peers and impose an additional minimum Tier 1 capital (and not only CET 1) amount grounded on accounting values. G-SIB banks have an additional buffer over the 3% value.

Tier 1 Capital (CET 1 + AT1)

$\geq 3\%$

1 Total assets (except derivatives values)

2 Off-Balance Sheet commitment (e.g. guarantees offered, financing commitments). Notionals are multiplied by a (arbitrary set) conversion factor.

3 Derivatives EAD measured though SA-CCR or OEM (IMM not allowed)

Pillar 1: Minimum required Liquidity – introduction to the Net Stable Funding Ratio (“NSFR”)

Regulators impose also (since 2019) to all banks a “ready-to-wear / one size fits all” to cover liquidity risk on the long term (also called funding risk). The idea is to ensure that the assets are refinanced with liabilities displaying a greater maturity than those of the assets to avoid having a cash outflow not covered by a cash inflows. We say that the assets ***require stable funding (“RSF”) whilst*** the liabilities offer an available stable funding **(“ASF”)**. The net stable funding ratio (ASF/RSF) should always be larger than 100% .

NSFR = $\dfrac{\text{Amount of available stable funding (ASF)}}{\text{Amount of Required Stable funding (RSF)}}$ **≥ 100%**

Example #1 (Pass – NSFR > 100%)

Asset	Liabilities	RSF	ASF
Loans : 100, bullet profile, 5 years	Sight retail deposits: 100	100 x 65%=65	100*95%=95
Cash : 5	Equity: 5	5 x 0%=0	5*100%=5

NSFR = $\dfrac{95+5}{65}$ **≥ 100%** ☑

Example #2 (Fail – NSFR < 100%)

Asset	Liabilities	RSF	ASF
Loans : 100, bullet profile, 5 years	Bonds: 100, bullet profile, 1 year	100 x 65%=65	100*50%=50 0
Cash : 5	Equity: 5	5 x 0%=0	5*100%=5

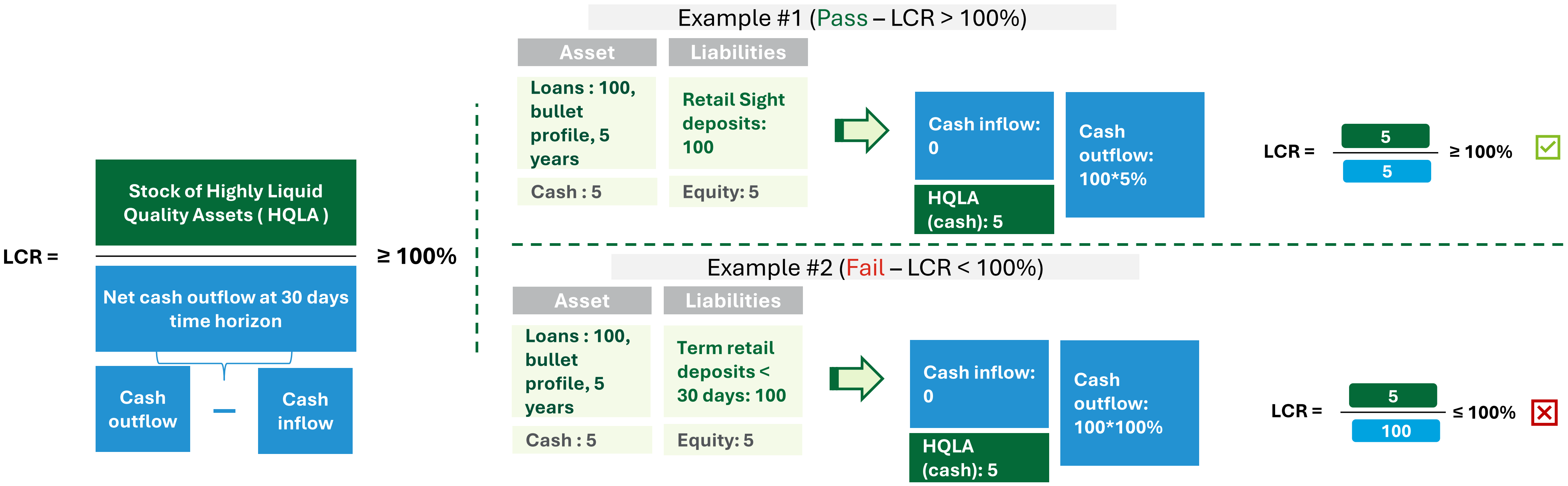
NSFR = $\dfrac{50+5}{65}$ **≤ 100%** ☒

- **The various weights (e.g. *0%, 65%, 90%, 100%)* are set by the regulatory requirements** and solely depend on the nature of the assets & liabilities. Computing the NSFR solely requires to map each accounting entry onto its corresponding regulatory weight. **No internal approach exists for the NSFR**.
- We observe that the sight deposits retail (even if they may contractually be withdrawn overnight) have a **95%** (and in some instances 90%– not detailed here) **ASF weight**, much favourable than a refinancing through a bond with a 1Y Maturity (**50%** weight). The weight have been set in a manner to not over penalize the funding strategy through retail deposits which is in essence the core role of the banking business.

Pillar 1: Minimum required Liquidity – introduction to the Liquidity Coverage Ratio (“LCR”)

Regulators impose also (since 2014) to all banks a “ready-to-wear / one size fits all” to cover liquidity risk on the short-term. The regulators require to screen all assets & liabilities (including commitments to receive or provide liquidity – and are off-balance sheet) with a contractual maturity **less than 30 days** and to multiply their accounting values by a fixed weight. **Eligible assets** are transformed into equivalent **cash inflows** whilst eligible liabilities are transformed into **cash outflows**.

The bank is required to permanently hold a portfolio composed of **Highly Liquidity Quality Assets** (e.g. Cash deposited in the central bank, Government Bonds from EU countries) to withstand the **net cash outflows. In other words, the banks must hold a certain amount of HQLA to face a <u>potential</u> bank run and avoid bankruptcy.**



- Net cash outflows is always positive as the regulators impose a floor (not detailed here) on the cash-inflows. In the example 2, the bank should seek for an immediate additional financing source to restore its LCR (and avoid a breach of regulation).
- The weight applied on the sight deposits amounts to 5%, this is quite significant as it is equivalent over 12 months as a cumulated cash-outflows of 60% (12x5%). Equity is never triggering an outflow.

Pillar 2: Overview – Internal Capital Adequacy Assessment Process (“ICAAP”)

Pillar 1 is a “one size-fits all” approach. However, as each bank is different (e.g. size, complexity), the Pillar 2 imposes to all banks to make “their own bespoke adjustments”, to eventually come up with a mandatory (positive) “add-on” on the Pillar 1 Capital (CET 1, AT1 or T2). This process is denoted as Internal Capital Adequacy Assessment (“ICAAP”).For instance: if the outcome of the ICAAP is an add-on of 0.5% on the CET1, **the bank should never have a level of CET 1 capital falling below 5% (4.5%+0.5%) of its RWA (otherwise it is no longer allowed to operate and the supervisor withdraws its banking license**.

Step 1 SELECT & RE-MEASURE ALL KEY RISKS FROM THE BANK’S VIEWPOINT (EXAMPLE)

	Risk category	Relevance assessment for bank X	Internal quantification in ICAAP**
ELEMENT 1 “Pillar 1 risks”	CREDIT RISK	Y	Y
	COUNTERPARTY RISK*	Y	Y
	MARKET RISK	Y	Y
	OPERATIONAL RISK	Y	N
ELEMENT 2 “Pillar 2 risks”	RESIDUAL RISK	Y	N, qualitative
	SECURITISATION RISK	Y	Y
	CONCENTRATION RISK	Y	Y
	Structural Interet Rate Risk	Y	Y
	FUNDING COSTS	Y	On-going
	REPUTATION RISK	Y	N, explained
	BUSINESS/STRATEGIC RISK	Y	On-going
	OTHER RISKS**	N**	N

*Any relevant risk that bank deem as applicable for its business

** Any relevant quantification method is possible, provided that it measures an UL over 1 year with a 99,9% quantile

Add-on = max (UL internally quantified - Pillar 1 UL ,0)

Step 2 Projection over 3 years of the strategic business plan

	Y+1		Y+2		Y+3	
	Balance Sheet ---------- RWA ----------- CAPITAL		Balance Sheet ---------- RWA ----------- CAPITAL		Balance Sheet ---------- RWA ----------- CAPITAL	
	CAPITAL RATIO(S)		**CAPITAL RATIO(S)**		**CAPITAL RATIO(S)**	
Solvency Ratio	≥ TC+(add-on)+CCB+CCyB	✓	≥ TC+(add-on)+CCB+CCyB	✓	≥ TC+(add-on)+CCB+CCyB	✓
	≤ TC+(add-on)+CCB+CCyB	✗	≤ TC+(add-on)+CCB+CCyB	✗	≤ TC+(add-on)+CCB+CCyB	✗
Leverage Ratio	≥ 3%	✓	≥ 3%	✓	≥ 3%	✓
	≤ 3%	✗	≤ 3%	✗	≤ 3%	✗

Step 3

Same as step 2 but including (internally designed) stress-scenarios affecting the total balance sheet, certain risks and hence RWAs.

Solvency Ratio	≥ 8%+add-on	✓
	≤ 8%+add-on	✗

Pillar 2: Overview – Internal Liquidity Adequacy Assessment Process (“ILA

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Pillar 2 on liquidity is conducted in the same vein as for the Capital to come up with a bespoke approach on liquidity. Banks are required to conduct their own internal assessment and come up with specific adjustments on the LCR & NSFR (either through add-ons on the minimum thresholds or review of the regulatory weights applied to compute those ratios). Institutions may also design additional liquidity metrics. This process is denoted as Internal Liquidity Adequacy Assessment (“ILAAP”).
However, in the EU, the outcome of the ILAAP cannot lead to mandatory add-ons on the LCR & NSFR (as opposed to the Capital) – the 3 steps below are applicable in the EU.

Step 1 – SELECT & RE-MEASURE ALL KEY LIQUIDITY RISKS FROM THE BANK’S VIEWPOINT (EXAMPLE)

	Risk category	Relevance assessment for bank X	Internal reassessment in ILAAP
ELEMENT 1 “Pillar 1 risks”	LCR	Y	Y
	NSFR	Y	Y
ELEMENT 2 “Pillar 2 risks”	CONCENTRATIONS	Y	N, qualitative
	AVAILABILITY IN CASE OF STRESS	Y	Y
	CONCENTRATION RISK	Y	Y
	INTERNAL LIQUIDITY METRIC* #1	Y	Y
	INTERNAL LIQUIDITY METRIC* #2	Y	N
	INTERNAL LIQUIDITY METRIC* #N	Y	N

** Any relevant quantification method / metric is possible

Step 2 – Projection over 3 years of the strategic business plan

	Y+1	Y+2	Y+3
	LCR / NSFR ---------- « Internal » LCR /NSFR ---------- Other internal designed metrics	LCR / NSFR ---------- « Internal » LCR /NSFR ---------- Other internal designed metrics	LCR / NSFR ---------- « Internal » LCR /NSFR ---------- Other internal designed metrics
Pillar 1	LCR ≥ 100% NSFR ≥ 100% ✓	LCR ≥ 100% NSFR ≥ 100% ✓	LCR ≥ 100% NSFR ≥ 100% ✓
	LCR ≤ 100% or NSFR ≤ 100% ✗	LCR ≤ 100% or NSFR ≤ 100% ✗	LCR ≤ 100% or NSFR ≤ 100% ✗
	Internal metrics & thresholds	Internal metrics & thresholds	Internal metrics & thresholds

Step 3

Same as step 2 but including (internally designed) stress-scenarios affecting any relevant liquidity risk. Even under-stress LCR and NSFR should remain above their minimum Pillar 1 thresholds.

Banking Union and Resolution in the EU

Banking Union in the EU since 2012 – 3 building blocks

Unique Supervisory Mechanism

- Ensure consistency of the supervision across the EU
- Ensure the resilience of the EU banking system
- Reinforce the integration and financial stability

EUROPEAN CENTRAL BANK

Banking Resolution

- Single Resolution Mechanism (SRM) under the leadership of the EU
- The SRM is articulated around the Single Resolution Board (SRB) and Single Resolution Fund (SRF)
- Aims at solving the potential banking crisis and failures with a lower cost for the taxpayers.

Single Resolution Board

Deposit Guranteed Scheme (DGS)

- Foster the level of confidence from the households towards banks
- Ensure that the savings (Households & Depositors) are not lost in case of bankruptcy.
- Ensure that the deposits & savings are paid back.

An overview of regulatory reporting – how supervisory bodies are informed ?

Supervisory Bodies

Regulatory Reporting (e.g. COREP, FINREP, SURFI)

Credit Institutions

Are entitled to take loans & deposits but also provide dealing on own account, underwriting.

≠

Investment Firms

Are not entitled to grant loans & take on deposits, but can run dealing on own account, underwriting, asset management

What?

Submission of certain financial, accounting risk and non-financial data. Financial institutions demonstrates their compliance to existing regulations and laws by providing these information.

Who ?

Credit institutions & investment firms are responsible for producing and delivering these reports.

To whom ?

To supervisory bodies such as the ACPR (Autorité de contrôle prudentiel et de résolution) and the ECB (European Central Bank)

What for?

To improve the functioning of the financial markets by ensuring appropriate, efficient and harmonized European regulation and supervision and creating a level-playing field within EU. To reduce the likelihood of financial crisis and contagion within the banking system.

Reporting or Disclosure ?

Reporting is private and granular information dedicated to the supervisory bodies (ex: COREP and FINREP are confidential), whilst a **disclosure** is a public information (ex: Pillar III). However, in some instances, subset of reportings may be disclosed into the Pillar III.

Main accounting & regulatory notions

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Here some examples of main regulatory reporting – These are becoming more and more data driven

Reporting Name	Description
COREP	**Common Report ("COREP")** is a reg. report introduced trough the CRD IV regulation. It presents the main risks (credit, market and operational), solvency & leverage ratios (capital ratios) and liquidity ratios (LCR and NSFR). The COREP's framework is considered as the main and most important regulatory report, it serves as a "risk identity card" and ensure that banks are in a capacity to operate. **Important:** small investment firms running certain typology of activities (outside trading or underwriting) have been subject to specific COREP reports since 2021 in accordance with IFR/IFD (Investment Firm Regulation / Directive).
FINREP	**Financial Report ("FINREP")** is more accounting related. It was also introduced trough the CRD IV regulation. All the accounts from the financial statements are presented using IFRS standards. It uses the ledger as a primary source of information in order to have granular data.
RUBA	**Reporting Unifié des Banques & Assimilés ("RUBA")** in force since January 31st,2022 (and replacing the Système Unifié de Reporting Financier **("SURFI"))** is a French reporting produced for the Banque de France and for the French banking supervisory body (ACPR). This reports mixes numerous kind of data : accounting data, statistical data used by the BdF to monitor the monetary and financial activity, non-financial data (e.g. headcounts, business locations).
ANACREDIT	Analytical Credit Datasets **("Anacredit")** reports, for each credit institution's counterparty (retail clients are excluded) the total exposure composed of Anacredit eligible instruments (derivatives are excluded) when equal to or greater than €25,000k. Specific information about Counterparties (e.g. nature, ultimate parent company), collaterals & guarantees are provided on a granular level. The ECB uses Anacredit as a tool to drive its monetary policy and macroprudential supervision.
Green Asset Ratio	The **Green Asset Ratio ("GAR")** is a newly established ratio covers asset and off-balance sheet exposures. This metric's objective is to make sure that bank's asset are aligned with a green taxonomy provided by the regulators, in other words it indicates how "green" is the Balance sheet.

Regulatory reporting – "Cheat sheet" – For FRENCH SPEAKERS

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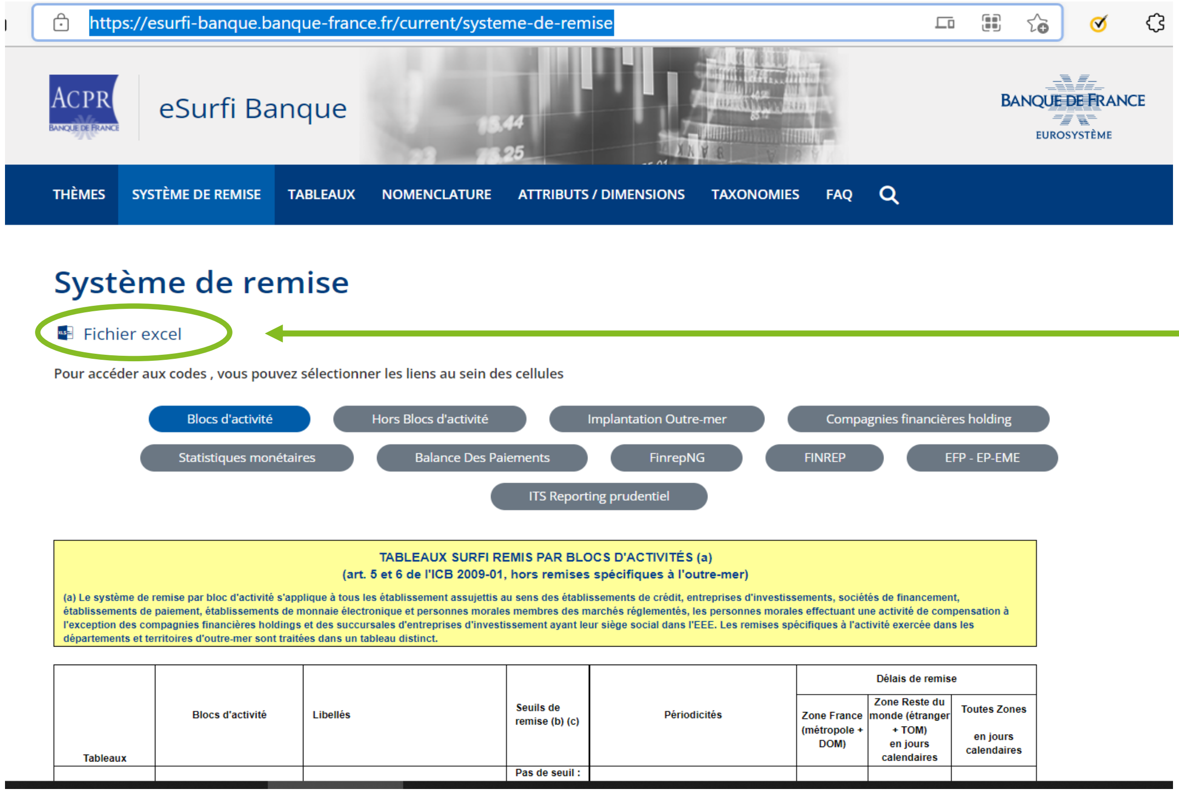
Utilising Esurfi as a core resource to look for any information when it comes to certain regulatory reports



Templates

Click here to have access to SURFI, FINREP and COREP Reports under Excel (including detailed handbook to populate those reports)

Where to find the regulatory reportings ?

Add to your favourites of your web browser : https://esurfi-banque.banque-france.fr/current/systeme-de-remise